

CURV

July 13, 2025

```
[1]: import pylab as pl
import numpy as np
from matplotlib.gridspec import GridSpec
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[2]: EF = 12.252300
tick_labels = ['$\Gamma$', 'X', 'R', 'Z', '$\Gamma$', 'M', 'A', 'Z']
tick_locs = [0, 0.691905, 1.694871, 2.386776, 3.389743, 4.368243, 5.371210, 6.
↪349711]
```

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<>:2: SyntaxWarning: invalid escape sequence '\G'
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C:\Users\ASUS\AppData\Local\Temp\ipykernel_3868\248528181.py:2: SyntaxWarning:
invalid escape sequence '\G'
    tick_labels = ['$\Gamma$', 'X', 'R', 'Z', '$\Gamma$', 'M', 'A', 'Z']
C:\Users\ASUS\AppData\Local\Temp\ipykernel_3868\248528181.py:2: SyntaxWarning:
invalid escape sequence '\G'
    tick_labels = ['$\Gamma$', 'X', 'R', 'Z', '$\Gamma$', 'M', 'A', 'Z']
```

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[17]: fig = pl.figure(figsize=(8, 6))
gs = GridSpec(2, 1, hspace=0.00)

# Load data files
databand = np.loadtxt('up/RuO2-bands.dat')
data1 = np.loadtxt('up/RuO2-curv.dat')
data2 = np.loadtxt('down/RuO2-curv.dat')

# Top subplot: bands
axes1 = pl.subplot(gs[0, 0])
pl.scatter(databand[:,0], databand[:,1]-EF, color='k', marker='.', s=1.0)
pl.xlim([0, max(databand[:,0])])
pl.ylim([-0.65, 0.65])
pl.plot([tick_locs[0], tick_locs[-1]], [0, 0], color='red', linestyle='--',
↪linewidth=1.0)
pl.xticks(tick_locs, tick_labels)
for n in range(1, len(tick_locs)):
    pl.plot([tick_locs[n], tick_locs[n]], [pl.ylim()[0], pl.ylim()[1]],
↪color='gray', linestyle='-', linewidth=0.5)
```

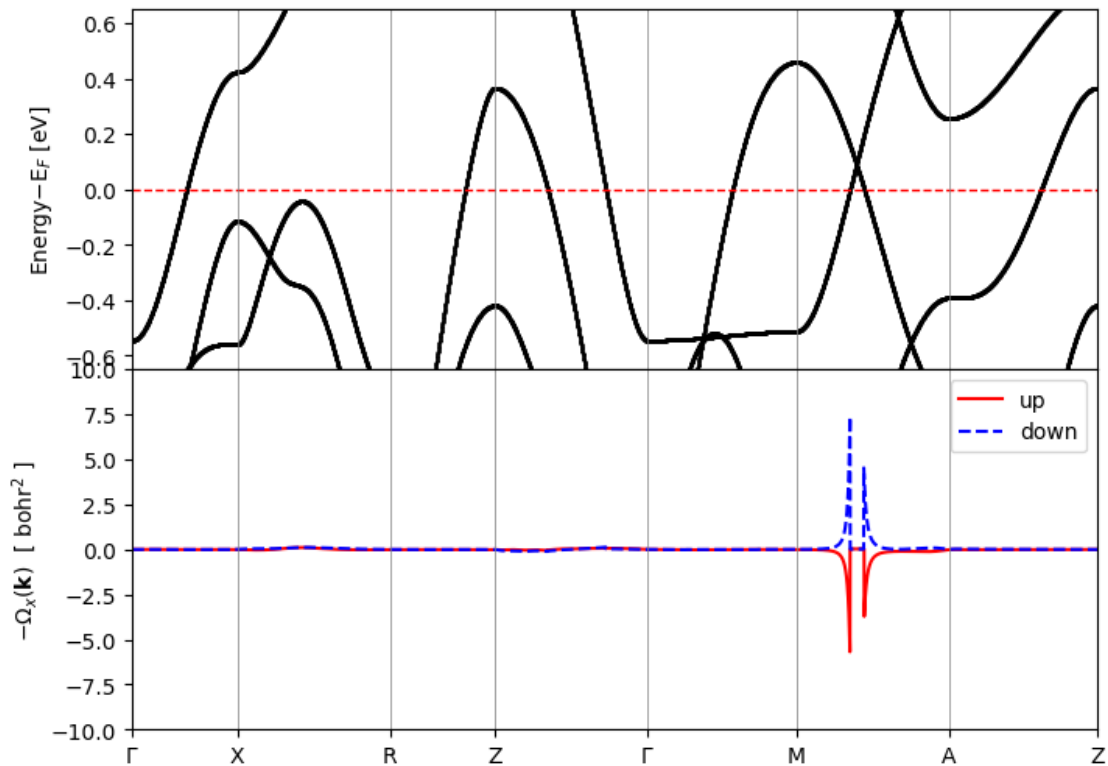
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pl.ylabel('Energy$-E_F$ [eV]')
pl.tick_params(axis='x', which='both', bottom=False, top=False,
    ↳labelbottom=False)

# Bottom subplot: curvature y
axes2 = pl.subplot(gs[1, 0:])
pl.plot(data1[:,0], data1[:,1], color='red', linestyle='-', label='up')
pl.plot(data2[:,0], data2[:,1], color='blue', linestyle='--', label='down')
pl.xlim([0, max(data1[:,0])])
#pl.ylim([min(data1[:,1])-0.025*(max(data1[:,1])-min(data1[:,1])), max(data1[:,
    ↳1])+0.025*(max(data1[:,1])-min(data1[:,1]))])
pl.ylim(-10,10)
pl.xticks(tick_locs, tick_labels)
for n in range(1, len(tick_locs)):
    pl.plot([tick_locs[n], tick_locs[n]], [pl.ylim()[0], pl.ylim()[1]],
    ↳color='gray', linestyle='-', linewidth=0.5)
pl.ylabel(r'$-\Omega_x(\mathbf{k})$ [ bohr$^2$ ]')

outfile = 'RuO2-bands+curv_y.pdf'
pl.savefig(outfile, bbox_inches='tight')
pl.legend(loc='upper right')
pl.show()

```



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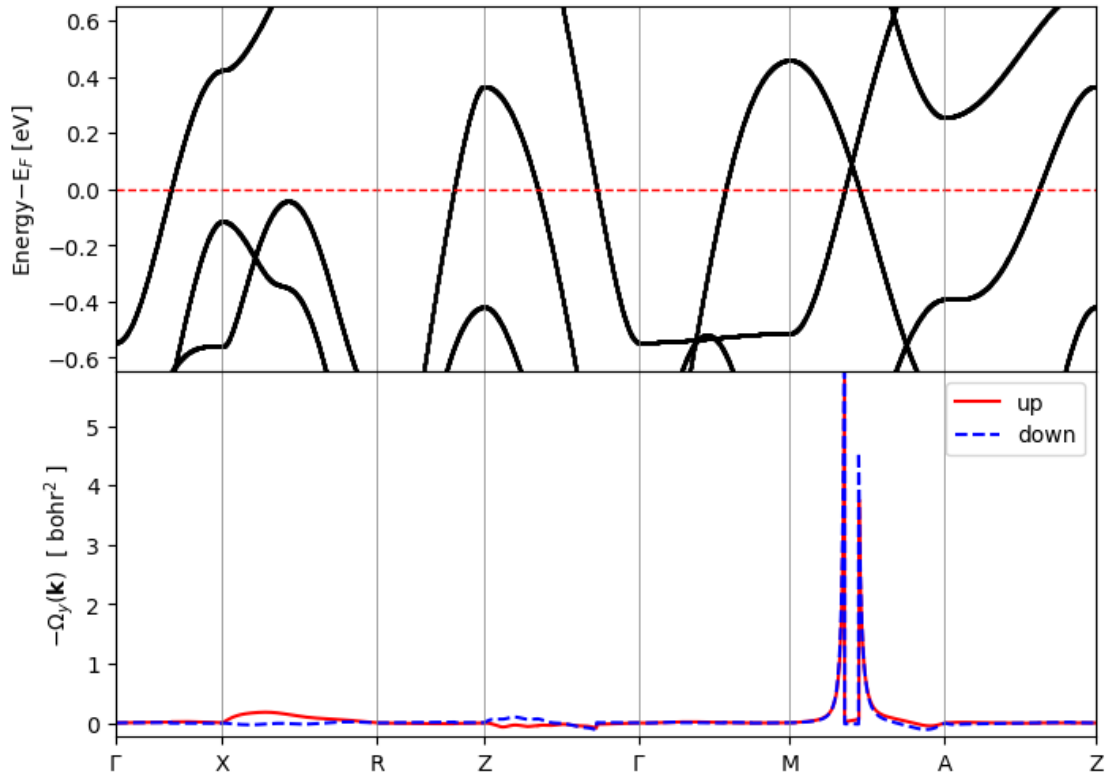
[15]: fig = plt.figure(figsize=(8, 6))
gs = GridSpec(2, 1, hspace=0.00)

# Load data files
databand = np.loadtxt('up/RuO2-bands.dat')
data1 = np.loadtxt('up/RuO2-curv.dat')
data2 = np.loadtxt('down/RuO2-curv.dat')
# Top subplot: bands
axes1 = plt.subplot(gs[0, 0:])
plt.scatter(databand[:,0], databand[:,1]-EF, color='k', marker='.', s=1.0)
plt.xlim([0, max(databand[:,0])])
plt.ylim([-0.65, 0.65])
plt.plot([tick_locs[0], tick_locs[-1]], [0, 0], color='red', linestyle='--',
        linewidth=1.0)
plt.xticks(tick_locs, tick_labels)
for n in range(1, len(tick_locs)):
    plt.plot([tick_locs[n], tick_locs[n]], [plt.ylim()[0], plt.ylim()[1]],
            color='gray', linestyle='-', linewidth=0.5)
plt.ylabel('Energy$-E$F$ [eV]')
plt.tick_params(axis='x', which='both', bottom=False, top=False,
        labelbottom=False)

# Bottom subplot: curvature y
axes2 = plt.subplot(gs[1, 0:])
plt.plot(data1[:,0], data1[:,2], color='red', linestyle='-', label='up')
plt.plot(data2[:,0], data2[:,2], color='blue', linestyle='--', label='down')
plt.xlim([0, max(data1[:,0])])
plt.ylim([min(data1[:,2])-0.025*(max(data1[:,2])-min(data1[:,2])), max(data1[:,2])+0.025*(max(data1[:,2])-min(data1[:,2]))])
plt.xticks(tick_locs, tick_labels)
for n in range(1, len(tick_locs)):
    plt.plot([tick_locs[n], tick_locs[n]], [plt.ylim()[0], plt.ylim()[1]],
            color='gray', linestyle='-', linewidth=0.5)
plt.ylabel(r'$-\Omega_y(\mathbf{k})$ [ bohr$^2$ ]')

outfile = 'RuO2-bands+curv_y.pdf'
plt.savefig(outfile, bbox_inches='tight')
plt.legend(loc='upper right')
plt.show()

```



```
[9]: fig = pl.figure(figsize=(8, 6))
gs = GridSpec(2, 1, hspace=0.00)

# Load data files
databand = np.loadtxt('up/RuO2-bands.dat')
data1 = np.loadtxt('up/RuO2-curv.dat')
data2 = np.loadtxt('down/RuO2-curv.dat')
# Top subplot: bands
axes1 = pl.subplot(gs[0, 0:])
pl.scatter(databand[:,0], databand[:,1]-EF, color='k', marker='.', s=1.0)
pl.xlim([0, max(databand[:,0])])
pl.ylim([-0.65, 0.65])
pl.plot([tick_locs[0], tick_locs[-1]], [0, 0], color='red', linestyle='--',
        linewidth=1.0)
pl.xticks(tick_locs, tick_labels)
for n in range(1, len(tick_locs)):
    pl.plot([tick_locs[n], tick_locs[n]], [pl.ylim()[0], pl.ylim()[1]],
            color='gray', linestyle='-', linewidth=0.5)
pl.ylabel('Energy-$E_F$ [eV]')
pl.tick_params(axis='x', which='both', bottom=False, top=False,
               labelbottom=False)
```

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# Bottom subplot: curvature z
axes2 = pl.subplot(gs[1, 0:])
pl.plot(data1[:,0], data1[:,3], color='red', linestyle='-', label='up')
pl.plot(data2[:,0], data2[:,3], color='blue', linestyle='-', label='down')
pl.xlim([0, max(data[:,0])])
pl.ylim([min(data1[:,3])-0.025*(max(data1[:,3])-min(data1[:,3])), max(data1[:,3])+0.025*(max(data1[:,3])-min(data1[:,3]))])
pl.xticks(tick_locs, tick_labels)
for n in range(1, len(tick_locs)):
    pl.plot([tick_locs[n], tick_locs[n]], [pl.ylim()[0], pl.ylim()[1]], color='gray', linestyle='-', linewidth=0.5)
pl.ylabel(r'$-\Omega_z(\mathbf{k})$ [ bohr$^2$ ]')

outfile = 'RuO2-bands+curv_z.pdf'
pl.savefig(outfile, bbox_inches='tight')
pl.legend(loc='upper right')
pl.show()

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